Roll No. .....

TCE–401

B. TECH. (CIVIL ENGINEERING) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

HYDRAULICS AND HYDRAULIC MACHINES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Determine the conditions for maximum discharge through a circular channel. (CO1)

(b) Find the rate of flow and conveyance for a rectangular channel 7.5 m wide for uniform flow at a depth of 2.25 m. The channel is having bed slope as 1 in 1000.Take Chezy's constant C = 55. Also state whether the flow is tranquil or rapid. (CO1)

(c) A wide rectangular channel carries a flow of 2.76 m^3/s per metre width, the depth of flow being 1.524 m. Calculate the minimum rise in the floor at a section required to produce critical flow condition. (CO1)

P. T. O.

2. (a) Sketch the surface profiles in the following cases : (CO2)

   (i) A mild slope changes to steep slope

   (ii) A Steeper slope changes to steep slope

   (b) Water flows in a triangular channel of side slope 1H : 1V and longitudinal slope of 0.001. Determine whether the channel is mild steep or critical when a discharge of 0.2 m^3/s flows through it. Assume Manning's $n = 0.015$. (CO2)

   (c) A sluice gate discharges water into a horizontal rectangular channel with a velocity of 10 m/s and depth of flow of 1 m. Determine the depth of flow after the jump and consequent loss in total head. (CO2)

3. (a) Explain the concept of boundary layer. Write short notes on various thicknesses of boundary layer. (CO3)

   (b) Air is flowing over a smooth plate with a velocity of 10 m/s. The length of the plate is 1.2 m and width 0.8 m. If Laminar boundary layer exist up to $Re = 2 \times 10^5$, find the maximum distance from the leading edge up to which laminar boundary layer exist. (CO3)

   (c) A kite weighing 9.8 N and having an area 1 m^2 makes an angle of 7° to horizontal when flying in a wind of 36 km/h. If pull on the string attached to the kite is 49 N and it is inclined to horizontal at 45°, Calculate the lift and drag coefficients. Take mass density for air = 1.2 kg/m^3.
   (CO3)

4. (a) Write a note on cavitation phenomena in a pump. (CO4)

(b) Derive an expression for the force exerted by a jet of water on a fixed vertical plate in the direction of the jet. (CO4)

(c) A centrifugal pump has an impeller of outer diameter 30cm. The vane tips are radial at the outlet. For a rotative speed of 1450 rpm, calculate the net head developed. Assume the manometric efficiency = 0.82.

(CO4)

5. (a) Define and explain hydraulic efficiency, mechanical efficiency and overall efficiency a turbine. (CO5)

(b) With the help of neat diagram explain the various elements of a hydroelectric power plant. (CO5)

(c) A Pelton turbine is to operate under a net head of 500 m at 420 rpm. If a single jet with dameter 18 cm is used, find the specific speed of the machine. Take $C_v = 0.98$, $\phi = 0.45$ and overall efficiency = 0.85.(CO5)

Roll No. .....

TCE–402

B. TECH. (CE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

STRUCTURAL ANALYSIS–I

Time : Three Hours

Maximum Marks : 100

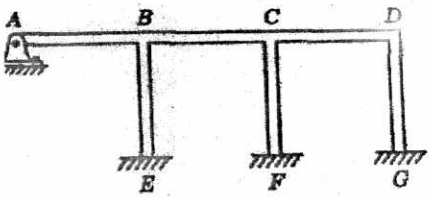
Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

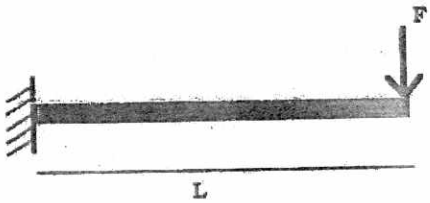
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) (i) Find the degree of static indeterminacy (DI) and Kinematic indeterminacy (KI or DOF) of the plane frame shown in fig. below : (CO1)

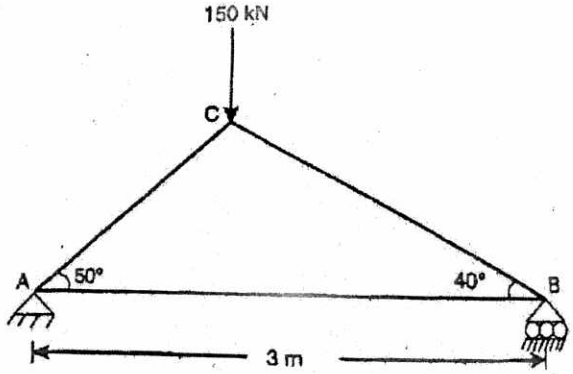


(ii) Draw the conjugate beam for the beam shown in fig. below :

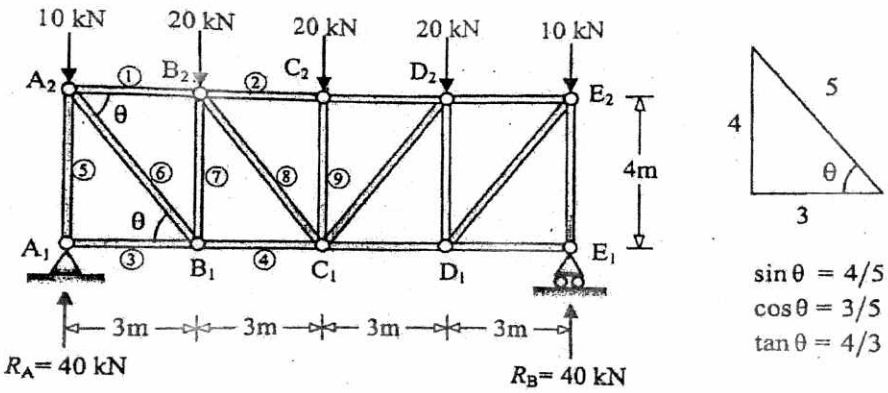


P. T. O.

(b) *Using tension coefficient method, determine all the forces of the truss shown in fig. below :* (CO1)

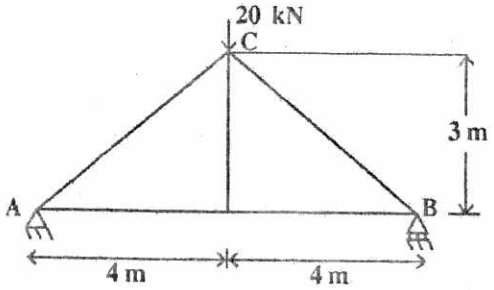


(c) Determine forces in members ②, ④ and ⑧ of the shown in fig. below by the method of sections : (CO1)



2. (a) Show the correspondence between original beam and conjugate beam with respect to supports, loading and responses. (CO2)

(b) For the given truss, find the horizontal and vertical deflection of the Point C using Castigliano's Theorem. Take AE = 10000 kN : (CO2)



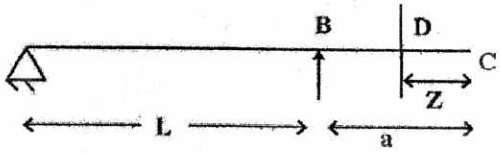
(c) State the following theorems with illustration : (CO2)

(i) Maxwell's reciprocal theorem

(ii) Castigliano's theorem

3. (a) State and illustrate the Muller-Breslau's principle with respect to drawing ILDs. (CO5)

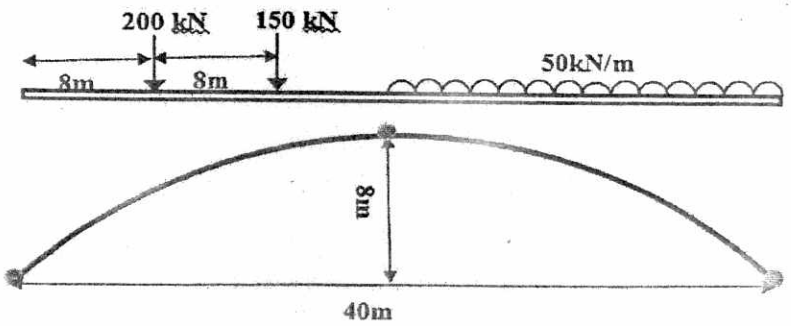
(b) Draw Influence line diagram for the shear force and bending moment at section D for the given beam (shown in fig.) : (CO5)



(c) Derive the condition for absolute maximum bending moment and shear force in a simply supported beam when a train of point loads crosses it. Also determine the position of the loads for causing the maximum bending moment and shear force at a particular section. (CO5)

4. (a) Explain the advantage of arch configuration. Explain the various types of common arches and their structural features. (CO5)

(b) A 3 hinged arch of span 40 m and rise of 8 m carries concentrated loads of 200 kN and 150 kN at distances of 8 m and 16 m from left end and a u.d.l. of 50 kN/m on the right half of the span. Find the horizontal thrust. (CO5)



(c) Formulate the horizontal thrust and bending moment at any section XX for a three hinged parabolic arch of span "l" and an UDL of w/m run over the entire span. (CO5)

5. (a) Draw the BMD for a propped cantilever beam of span 6 m with UDL of 10 kN/m over its entire span. (CO5)

(b) State the *three* moment area theorems and illustrate them through any example. (CO5)

(c) Determine the fixed-end moments when a span "L" is loaded by UDL of 'w'/m run over the entire span and a central point load P. (CO5)

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Roll No. .....

TCE–403

B. TECH. (CIVIL ENGG.) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

ADVANCED SURVEYING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the various classification of triangulation systems. (CO1)

(b) From a satellite station S, 14 metres to the west of the main station A, the following angles were measured :

Angle BSC = 68°26'36"

Angle CSA = 32°45'48"

The stations S and C are to the opposite sides of the line AB. Calculate the correct angle BAC if the lengths AC and AB are 5678 m and 1441 m respectively. (CO1)

(c) Explain method of extension of base line by prolongation method.(CO1)

P.T.O.

2. (a) Calculate the standard error of the volume of a box whose sides a, b and c have the values and standard error as under : (CO2)

$a = 3 \pm 0.02$ m

$b = 4 \pm 0.01$ m

$c = 3 \pm 0.02$ m

(b) Explain various law of weights.

(c) The following are the observed values of an angle : (CO2)

Angle	Weight
40°20'20"	2
40°20'18"	2
40°20'19"	3

(i) Probable error of single observation of unit weight

(ii) Probable error of weighted arithmetic mean

(iii) Probable error of single observation of weight 3

3. (a) Two tangents intersect at a chainage of 1320.50 m having deflection angle of 24°. If the radius of the curve to be laid out is 275 m, calculate the Length of the curve, Tangent length, Length of the long chord, Apex distance, versed sine of curve (Mid ordinate) and Chainage of point of commencement and point of tangency. (CO3)

(b) Define transition curve. Describe the advantages of a transition curve. Explain requirements of a transition curve. (CO3)

(c) Define super-elevation. What is the need of super-elevation and how it is determined ? Explain various types of vertical curve with neat sketches. (CO3)

4. (a) Explain the engineering project surveys, reconnaissance, preliminary and location surveys for highways in detail. (CO4)

(b) Describe the purpose of field astronomy. Define the following terms : (CO4)

(i) Celestial sphere

(ii) Equator

(iii) Latitude

(iv) Longitude

(v) Celestial horizon

(vi) Zenith

(vii) Nadir

(viii) Declination Circle

(c) Define hydro-graphic surveying. Explain the application and use of hydro-graphic surveying. Explain sounding in hydro-graphic surveying. (CO4)

5. (a) A line AB lies on a terrain having an average elevation of 500 m above means sea level. It appears to be 10.26 cm on a photograph for which focal length is 28 cm. The same line measures 3.28 cm on a map which is to a scale of 1/50,000. Calculate the flying altitude of the aircraft above mean sea level when the photograph was taken. (CO5)

(b) Explain the advantages and disadvantages of remote sensing. Explain the applications of remote sensing. (CO5)

(c) Write a short note on Geographic Information System (GIS). (CO5)

Roll No. .....

TCE–404

B. TECH. (CIVIL ENGINEERING)
(FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

CONCRETE TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) IS code 456:2000 and IS 10262:2019 will be provided.

1. (a) Describe the hydration reaction of important Bogue compounds indicating the products of hydration. (CO1)

(b) What is meant by grading of aggregates ? Explain in detail. (CO1)

(c) Explain setting time of cement and factors affecting setting time of cement. (CO1)

2. (a) Explain about different methods to measure workability of concrete. (CO2)

P. T. O.

(b) Explain the maturity concept for strength development of concrete. (CO2)

(c) Explain the following with reference to the properties of fresh concrete : (CO2)

(i) Segregation

(ii) Bleeding.

3. (a) What are the various factors affecting the compressive strength of concrete ? (CO3)

(b) Explain the various factors affecting shrinkage of concrete. (CO3)

(c) Explain the phenomenon of freezing and thawing in concrete and elaborate the steps taken to mitigate this durability issue in the concrete. (CO3)

4. (a) Carry out the following concrete mix design (As per IS 10262 : 2019)

Grade designation : M30 : (CO4)

- Type of cement : OPC 43 grade conforming to IS 269
- Type of mineral admixture : GGBS conforming to IS 16714
- Maximum nominal size of aggregate : 20 mm
- Minimum cement content and maximum water-cement ratio to be adopted and/or
- Exposure condition as per Table 3 and Table 5 of IS 456 : Severe (for reinforced concrete)
- Workability : 150 mm (slump)
- Method of concrete placing : Pumping
- Degree of supervision : Good

- Type of aggregate : Crushed stone angular aggregate
- Maximum cement (OPC content) : As per IS 456
- Chemical admixture type : Super plasticizer : normal

TEST DATA FOR MATERIALS

- Cement used : OPC 43 grade conforming to IS 269
  - o Specific gravity of cement : 3.15
- GGBS : Conforming to IS 16714
- Specific gravity of GGBS : 3
- Chemical admixture : Super plasticizer conforming to IS 9103
- Specific gravity (at SSD condition) of
  - o Coarse aggregate : 2.74 (based on saturated surface dry condition)
  - o Fine aggregate : 2.65 (based on saturated surface dry condition)
  - o Chemical Admixture : 1.145
- Water absorption
  - o Coarse aggregate : 1.5 percent
  - o Fine aggregate : 1.0 percent
- Moisture content of aggregate [per IS 2386 (Part 3)]
  - o Coarse aggregate : Nil
  - o Fine aggregate : Nil
- Sieve analysis
  - o Coarse aggregate : Conforming to Table 7 of IS 383
  - o Fine aggregate : Conforming to grading Zone III of Table 9 of IS 383

Provide the mix design quantity per m^3 of concrete.

(b) Define the term "Mix Design of Concrete" and explain its significance. (CO4)

(c) Explain the mix design procedure of concrete as per IS Code Method.

5. (a) Explain in brief the functions and mechanism of the "Super plasticizers" to be added in the concrete. (CO5, CO6)

(b) Write the purpose of mineral admixture. State any *four* mineral admixtures used in concrete. (CO5, CO6)

(c) Write the purpose of chemical admixture. State any *four* chemical admixtures used in concrete. (CO5, CO6)

Roll No. .....

TCE–405

B. TECH. (CIVIL ENGINEERING) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

GEOTECHNICAL ENGINEERING—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by residual soil and transported soil ? What are the *three* basic factors which influence the characteristics of a transported soil ? What factors determining the characteristics of a residual soil ? (CO1)

(b) Differentiate the following : (CO1)

(i) Dry unit weight and unit weight of soil solids

(ii) Saturated unit weight and Submerged unit weight

(iii) Undisturbed soil and Remolded soil

(iv) Sensitivity and Activity

(v) Liquid limit and Shrinkage limit

P. T. O.

(c) The following test results were obtained for two soil samples :

Parameters	Soil A	Soil B
% passing 75 micron	3	82
% passing 4.75 mm	97	98
D_{10}	0.11 mm	–
D_{30}	0.43 mm	–
D_{60}	1.18 mm	–
W_L	–	64
W_P	–	18

Classify the two soils as per BIS soil classification system. (CO1)

2. (a) What is a Flow Net ? What are the properties of Flow Net ? Explain the uses of a Flow Net. (CO2)

(b) Give *five* examples of Geotechnical Problems where knowledge of permeability is essential. Differentiate seepage velocity with superficial velocity. (CO2)

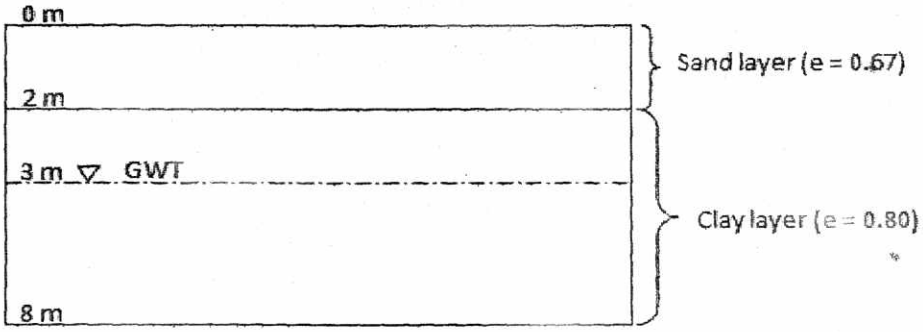
(c) For the given soil profile, plot the variation of total (vertical) stress, pore water pressure and effective (vertical) stress with depth for the following cases :

(i) For the case shown in the figure GWT is at 3.0 m depth. The soil above the GWT is dry.

(ii) GWT is at the ground surface.

(iii) GWT is 2 m above the ground surface.

Assume G_s=2.65 and $\gamma_w = 10\ kN/m^3$. (CO2)



3. (a) Define the following terms : (CO3)
   (i) Relative Density
   (ii) Relative Compaction
   (iii) ZAV line
   (iv) Optimum Moisture Content (OMC)
   (v) Maximum Dry Density (MDD)

   (b) Give an account of the differences in soil characteristics, such as, structure, permeability, shear strength, compressibility and swelling when it is compacted on the dry side of OMC and wet side of OMC. (CO3)

   (c) Following are the results of a compaction test : (CO3)

Weight of mould + wet soil (N)	Water content (%)
29.25	10
30.95	12
31.50	14.3
31.25	16.1
30.70	18.2

Volume of the mould = 1000 cc, Weight of the mould = 10 N and specific gravity of solids = 2.70.

(i) Plot the compaction curve showing the optimum moisture content and miximum dry density.

(ii) Plot the ZAV line and 80% saturation line.

(iii) Determine the degree of saturation at MDD.

(iv) Suggest the range of moisture content to be kept in the field to achieve RC equal to at least 90%.

4. (a) Discuss the following : (CO4)

(i) Normally consolidated and over consolidated soil

(ii) Compression Index and Recompression Index

(iii) Pore pressure and excess pore pressure

(iv) Primary Consolidation and Secondary Consolidation

(v) Single Drainage and Double Drainage condition

(b) How is consolidation different from compaction ? Discuss the factors that affect the compressibility of soil either in terms of amount of compressibility or its time rate. (CO4)

(c) A normally consolidated saturated clay stratum 2.5 m thick lies above the impervious layer and below a pervious sand layer. The compression index of the clay layer is 0.35 and its coefficient of permeability value 4.5×10^{-4} mm/sec. Its void ratio at the initial effective stress condition of 150 kPa was 1.95. Calculate the change in the void ratio due to additional stress of 80 kPa. Also, calculate the time (seconds) required for 50 % consolidation to occur. (CO4)

5. (a) Discuss the following : (CO5)

(i) Isobar and pressure bulb

(ii) Line load and Strip load

(iii) Peak and Residual shear strength

(iv) Critical Void Ratio

(v) Elasto-Plastic Material

(b) For a sedimentary soil deposit, which solution is more appropriate – Boussinesq or Westergaard ? Qualitatively, compare Boussinesq's coefficient (K_B) with Westergaard's coefficient (k_W). (CO5)

(c) For samples of silty clay, the cohesion and angle of internal friction from shear box test were obtained as cohesion c = 600 kPa and ϕ = 15°. If specimen of same soil is subjected to untrained triaxial test with confining stress of 300 kPa, calculate the deviator stress at which failure will be expected. (CO5)

Roll No. .....

TCE–406

B. TECH. (CE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

ENGINEERING GEOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is atmosphere ? Briefly explain and draw the structure of the atmosphere. (CO1)

(b) What is mineral ? Discuss any *five* physical properties of a mineral. (CO1)

(c) Discuss in detail about the importance of geology in Civil Engineering. (CO1)

2. (a) What are concordant and discordant bodies ? Briefly explain various concordant and discordant bodies. (CO2)

(b) What are metamorphic rocks ? Classify metamorphic rocks. (CO2)

P. T. O.

(c) Discuss the engineering properties of rocks. Also mention *five* examples each of igneous, sedimentary and metamorphic rocks. (CO2)

3. (a) What are folds and folding ? Discuss about parts of fold with the help of a neat diagram. (CO3)

(b) What are True and Apparent Dip ? Also discuss the types of Dip with the help of a neat diagram. (CO3)

(c) Discuss about the causes and classification of joint in rocks. (CO3)

4. (a) Define intensity of an earthquake. Also discuss the modified Mercalli intensity scale. (CO4)

(b) What is underground water and what are the different sources of underground water ? (CO4)

(c) Discuss the zonal classification for seismic zones in India. (CO4)

5. (a) Define tunnel and discuss about geological investigation for site selection of tunnel. (CO5)

(b) Discuss about the methods of geo-physical explorations. (CO5)

(c) Discuss the classification of dams. (CO5)

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